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PALAEOMAGNETISM OF KETILIDIAN METAMORPHIC ROCKS OF SW GREENLAND, AND 1850–1600 m.y. APPARENT POLAR MOVEMENTS

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Proterozoic supracrustal rocks of southwest Greenland and amphibolite dykes intruding the basement possess a thermal remanent magnetisation acquired during slow regional uplift and cooling between 1800 and 1600 m.y. following the Ketilidian mobile episode. Most samples from amphibolite dykes (mean palaeomagnetic pole 214°E, 31°N) possess a stable remanence associated with development of hematite during regional thermal metamorphism. Metavolcanics from the eastern part (eight sites, palaeomagnetic pole 230°E, 60°N, A_{95} = 15°) and western part (twelve sites, 279°E, 59°N, A_{95} = 17°) of Arsůk Island have magnetisations postdating folding and are related to K–Ar ages dating regional cooling (1700–1600 m.y.); magnetic properties are highly variable and partially stable remanence resides predominantly in pyrrhotite.

These results agree in part with other palaeomagnetic results from the northern margin of the same craton, and currently available palaeomagnetic results assigned to the interval 1850–1600 m.y. are evaluated to define apparent polar wander movements. Two large polar movements are recognised during this interval with the possibility of a third at ca. 1800 m.y. It is concluded that apparent polar wander movements in Proterozoic times are most accurately described in terms of closed loops.

1. Introduction

Notwithstanding the accumulation of considerable Precambrian palaeomagnetic data is recent years, the definition of apparent polar wander (a.p.w.) movements is still often rendered difficult by the great periods of time involved, and the problems of assigning ages to the magnetisations present in Precambrian rocks. In some cases it is likely that polar movements have occurred during the intervals of deposition of certain sediments (e.g., du Bois 1962; Jones and McElhinny 1967), the intervals of cooling of igneous bodies (e.g., Piper 1976a), and the slow cooling of metamorphic terrains (Morgan, 1976), although the interpretation of the evidence may be subjective and depend on the assessment of the research worker concerned. However, a.p.w. paths show a progressive evolution to greater complexity as more data have been forthcoming (e.g., Spall, 1971; Irving and Park, 1972; McGlynn et al., 1975; Piper, 1976b) so that they must eventually approach the complete path of the pole relative to the continent or continents concerned. It is important therefore that too much interpretation should not be placed on a.p.w. paths as defined because they depend critically on the available data; this point has often been recognised by workers who have defined overall "trends" in the data (e.g., Spall, 1971).

As a consequence of the long time intervals involved, it is unlikely that any one Precambrian region is going to provide a complete history of apparent polar movement. This paper reports palaeomagnetic results from metamorphic rocks of southern Greenland

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which, together with other results from further to the north on the same craton, show that the history of a.p.w. movements between 1850 and 1600 m.y. is considerably more complicated than would be inferred from present results from the adjacent Superior craton of North America.

2. Geology and age

The Ketilidian mobile belt in southern Greenland lies along the southern margin of a major block of Archaean (>2500 m.y.) gneisses (Bridgwater et al., 1973). This block comprises a migmatitic infrastructure overlain by a sedimentary and volcanic supracrustal suite including the Arsûk volcanic group of this study; most of the outcrop of this latter suite has experienced a complex tectonic history within the Ketilidian belt. Van Breemen et al. (1974) date the climax of metamorphism in this belt at 1840 ± 25 m.y. (U–Pb discordia age) and infer that the peak of metamorphism was of short duration and completed by 1800 m.y. Widespread and post-tectonic rapakivi-type intrusive episodes are recorded by U–Pb and Rb–Sr ages of 1780 ± 20 m.y. (van Breemen et al., 1974) and 1786–1774 m.y. (Bridgwater et al., 1973), and final uplift by Rb–Sr and K–Ar mineral ages from Sanerutian granites of 1700–1600 m.y. (Bridgwater, 1965; Larsen, 1971; Larsen and Møller, 1968; van Breemen et al., 1974).

The Arsûk group includes 4,200 m of pillow lavas, massive lavas and tuffs which overlie the sedimentary Ikerasarssuk group. Muller (1974) divides the Arsûk group into a dark series separating upper and lower light-coloured series, and sites *1–12* of this study form a stratigraphic section through the upper part of the lower, the middle and the lower part of the upper series, while sites *13–20* are entirely within the basal part of the lower series. The regional distribution of these palaeomagnetic sites is given in Fig. 1.

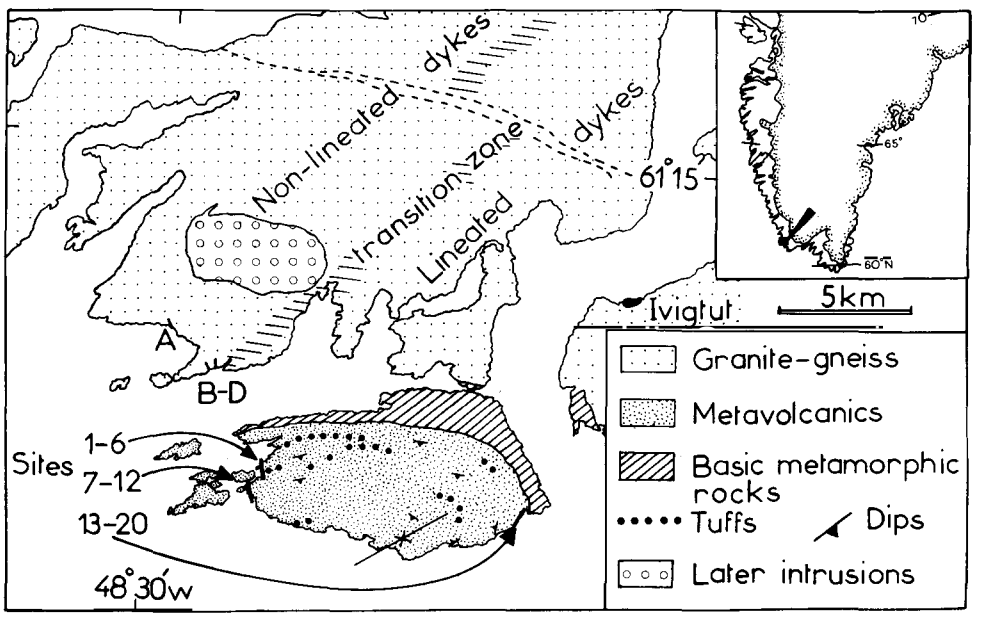


Fig. 1. Simplified geological map of the Ivigtut region, southwest Greenland (location is given on the inset map) showing the sampling localities for this study. The granite-gneiss is part of a major Archaean block (Bridgwater et al., 1973) and in the northern part of this region is cut by many dykes converted by a regional thermal metamorphic event to amphibolites. The duration of this event increased progressively from NW to SE in which direction the dykes have been progressively mobilised and reoriented across the transition zone indicated here after Bondesen and Henriksen (1965).

Feldspar microlites in the volcanics are heavily sericitised and actinolite amphibole, chlorite and epidote have entirely replaced any primary ferromagnesian components which may originally have been present; Muller (1974) reports some ilmenite, notably in the middle series. Secondary calcite, quartz and sulphides occur to a variable extent in all members both in the groundmass and as veinlets. Three phases of deformation are recognised in those rocks. The first produced a regional schistosity during greenschist facies metamorphism and NNE–SSW recumbent folds, the second produced a strain-slip cleavage associated with ESE-trending folds, and the third produced N–S-trending structures. The strain-slip cleavage of the second phase is well developed in the massive volcanics but present to some extent in all the pillowed units; metamorphism, increasing towards amphibolite facies in the lower members of the group, took place between the first and third stages of metamorphism.

In view of the complex tectonic and metamorphic history, any palaeomagnetic study must seek to establish the age of ancient components of magnetisation. This is facilitated on Arsûk Island because the supracrustal suite is exposed here as a syncline where sampling can be concentrated on two limbs of markedly differing strike and dip (Fig. 1). In the west sites *1–12* are sampled from volcanic units dipping 50–60° in a direction between 160 and 165°E and in the east sites *13–20* dip at an angle of 40° in a direction between 270 and 285°E.

Sites *A–D* are from dolerite dykes intruding the migmatitic infrastructure, and were metamorphosed by a regional heating event the effect of which increased from NW to SE (Bondesen and Henriksen, 1965) to produce recrystallisation of plagioclase and hornblende at temperatures above 400°C. Bondesen and Henriksen regard the effects of the metamorphism as a consequence of duration rather than grade of the metamorphic event, and the samples of this study are non-lineated but lie close to the zone where the dykes become lineated and concordant with structures in the basement (Fig. 1; Oen Ing Sven, 1962; Bondesen and Henriksen, 1965).

3. Palaeomagnetic results

All samples used in the study were collected in the field with a portable two-stroke drill, and oriented while still attached to the outcrop using sights on the sun, with the exception of some samples in the group of sites *13–20* which were oriented by magnetic compass in conjunction with topographic sights. Samples were measured in the laboratory using a parastatic magnetometer and progressively demagnetised by alternating fields (a.f.) in increasing steps of 50 Oe. Sample location data and principal magnetic properties are summarised in Table I.

Sites *A, C* and *D* from amphibolite dykes exhibit slight but systematic fall in moments and movement of magnetic vectors to shallower inclinations with progressive treatment (Fig. 2). Sites *C* and *D* show larger initial movements which may represent removal of low coercive force components acquired in the present geomagnetic field, but otherwise movements are small and systematic to the limits of treatment (800 Oe), although some samples from site *A* exhibit slightly more erratic behaviour above 200 Oe. Site *B* is much weaker than the other sites (Table I) and also less stable (Fig. 2); movement of magnetic directions is not systematic and not completely stable over any part of the range of treatment.

It is difficult to systematise the palaeomagnetic properties of the Ketilidian metavolcanics but a feature which is common to all sites is a large within-site variation in remanent intensity (Table I). The range of intensity between the *5–7* samples from each site is typically between one and two orders of magnitude. We infer that this is a function of amount, rather than type, of magnetic mineral content for three reasons. Firstly, the coercivities of strong and weak samples are comparable; secondly, variations in intensity tend to be reflected in variations in magnetic susceptibility (Table I); thirdly, there appears to be no general relationship between magnetic intensity and stability of directions, many weak samples are equally as stable as, and demagnetise in a similar way to, strong samples.

Several characteristics appear to apply to all of the sites *1–12*: some samples exhibit erratic change in moment with progressive a.f. treatment, and others exhibit regular fall in moments (Fig. 3), and although the former do not generally show less stability than the latter, the samples with higher coercivity are always more stable than those with lower coercivity. Samples of higher coercivity show highest stability of magnetic directions in the first 200 Oe of treatment; above this movements are larger and more erratic (Fig.

TABLE I

Location data and magnetic properties of samples from SW Greenland

Site	Location	Intensity of magnetisation (10^{-5} e.m.u./g)	Volume susceptibility (10^{-4} G/Oe)
	Section through 640 m of supracrustal sequences at: 48°25'W, 61°8'–61°9'N.		
1		0.06– 0.54	0.25 – 0.80
2		0.10– 1.25	0.10 – 0.73
3		0.19– 0.61	0.36 – 1.30
4		0.08– 1.27	0.0012– 0.0083
5		0.13– 2.44	0.34 – 1.03
6		0.09– 2.22	0.42 – 1.08
7		0.05– 0.17	0.45 – 0.85
8		0.13– 0.79	0.36 – 0.85
9		1.24–12.8	1.2 – 1.4
10		0.14– 1.82	0.38 – 0.90
11		0.11– 1.54	7.7 –16.0
12		0.08– 1.88	0.75 – 1.28
	Section through 380 m of supracrustal sequence at: 48°14'W, 61°8'N		
13		0.04– 1.45	0.81 – 1.18
14		0.05– 0.71	0.41 – 0.92
15		0.08– 3.69	0.62 – 1.07
16		0.87– 8.11	0.77 – 1.32
17		0.21– 4.98	0.74 – 0.87
18		0.04– 1.33	0.67 – 0.83
19		0.07– 9.66	0.81 – 1.41
20		0.07– 2.14	0.65 – 1.04
A	NNW dyke, 48°29'W, 61°11'N	0.38– 0.88	0.48 – 1.17
B	NNW dyke, 48°28'W, 61°10.5'N	0.03– 0.11	0.32 – 0.79
C	NE dyke, 48°28'W, 61°10.5'N	1.06– 6.62	0.26 – 1.92
D	NE dyke, 48°27'W, 61°10.5'N	0.10– 2.67	0.66 – 1.29

3) so that it did not generally prove worthwhile to demagnetise samples above 450 Oe. Of the second group of sites, *13*, *14* and *15* illustrate systematic fall in moment over the first 100–200 Oe of treatment followed by erratic changes in moment, and it is with these erratic changes that directions of magnetisation first show high instability. Partially stable site *16*, stable sites *17* and *18*, and poorly stable sites *19* and *20* exhibit comparable behaviour (Fig. 3) with magnetic moments and directions becoming unstable at higher or lower fields of treatment; in this group there appears to be no correlation between coercivity and stability. A comparison of the properties of the two groups of sites (*1–12* and *13–20*) demonstrates that it is impossible to generalise about properties of the whole collection: the two groups are quite dissimilar and sites *13* and *14*, for example, exhibit similar erratic fall in moments and ranges of NRM intensity to sites *4* and *5* although the latter two sites show very much higher stability of directions than the former. We have no explanation for this difference but suggest that it may relate to a complex acquisition of magnetic moments during slow regional cooling discussed in the next section.

Sample directions were selected from the most stable range of behaviour during demagnetisation with successive direction changes of 20° or less and combined to derive the site mean directions listed in Table II. In all cases this most stable range is within the first 100 or 200 Oe of treatment and it has not proved possible to derive concordant directions from demagnetisation results in higher fields. Because the combined statistics involve data from high palaeolatitudes, the ovals of confidence about the mean poles are rela-

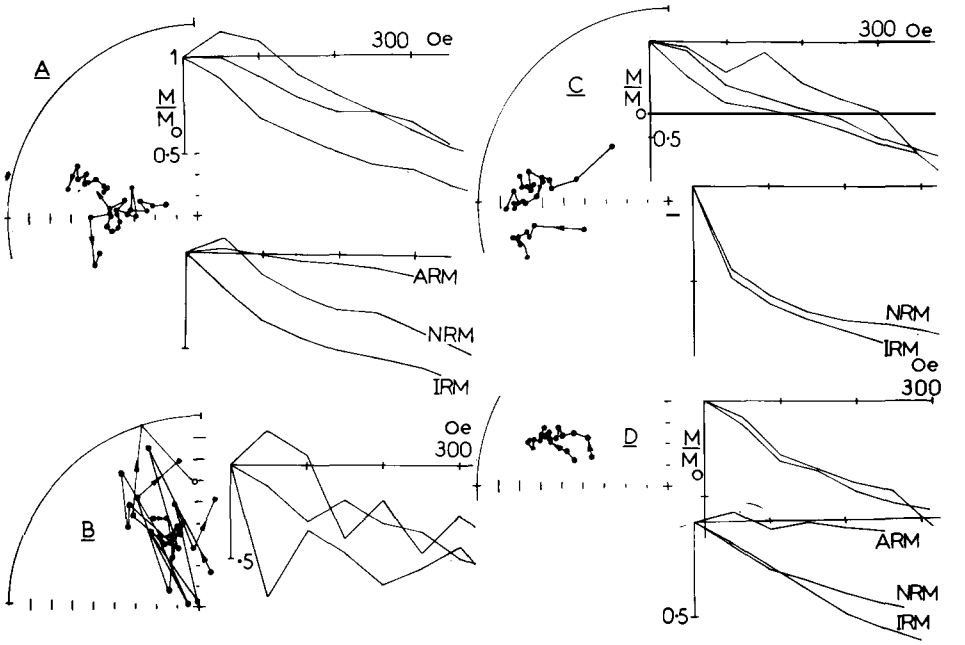


Fig. 2. Behaviour of typical samples from the amphibolite dykes showing change in direction with progressive a.f. demagnetisation, demagnetisation characteristics (M/M_0 vs. field in Oe), and demagnetisation characteristics of NRM's, IRM's and ARM's. Polar stereographic projections; solid symbols are lower-hemisphere plots and open symbols are upper-hemisphere plots.

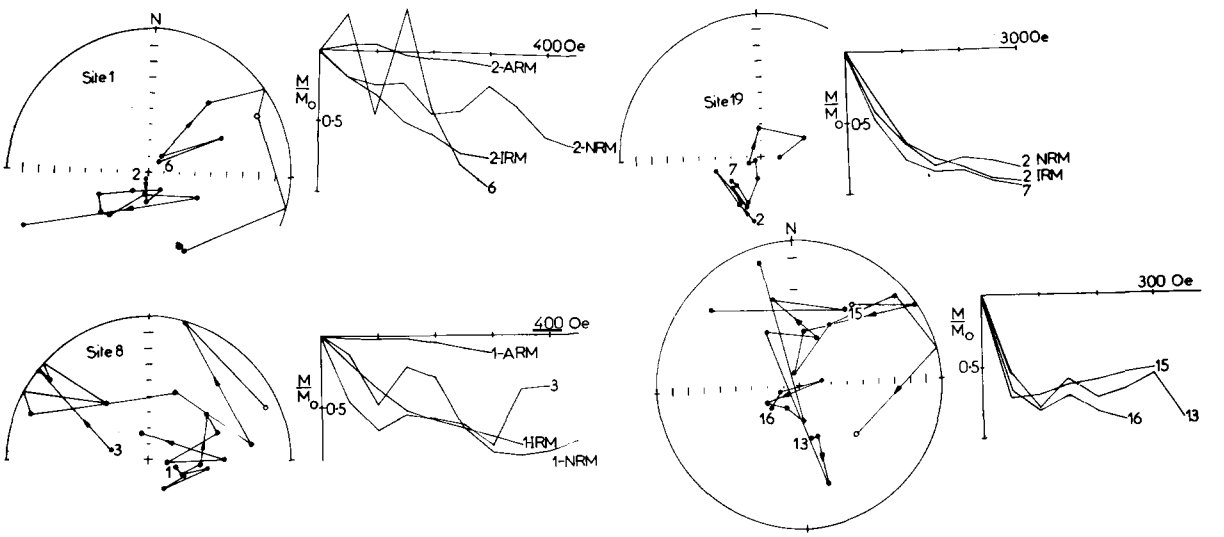


Fig. 3. Behaviour of some typical samples of Ketilidian metavolcanics with progressive a.f. demagnetisation. The numbers refer to sample numbers within individual sites.

TABLE II

Metamorphic rocks, SW Greenland

(a) Palaeomagnetic results after alternating field cleaning

Site	N	I (°)	D (°)	R	α_{95} (°)	Virtual geomagnetic pole	
						°N	°E
(i) Ketilidian metavolcanics, western part of Årsuk Island:							
1	4	82.3	229.2	3.72	30	50	294
2	6	83.2	47.3	5.43	24	68	339
3	6	74.8	221.4	5.14	30	37	288
4	6	66.9	287.1	3.77	30	49	239
5	4	60.1	321.5	3.34	49	60	199
6	6	77.7	169.7	4.46	30	38	317
7	5	54.3	283.1	4.53	28	36	230
8	6	81.1	21.0	4.44	45	76	338
9	6	74.1	289.0	5.74	16	57	252
10	6	79.6	311.0	5.05	32	69	266
11	5	78.5	208.4	4.54	28	41	298
12	5	75.1	143.8	4.49	29	36	332
(ii) Ketilidian metavolcanics, eastern part of Årsuk Island:							
13	6	66.8	334.9	5.20	30	72	193
14	6	61.3	268.3	4.83	37	35	247
15	7	57.2	315.8	5.40	36	54	202
16	7	69.9	273.3	6.44	19	44	250
17	7	83.4	345.4	6.91	7	74	300
18	6	61.8	311.0	5.88	10	56	212
19	7	69.1	278.9	6.49	18	48	248
20	6	64.9	341.9	5.51	22	72	176
(iii) Amphibolite dykes:							
A	6	55.2	279.4	5.86	12	35	233
B	4	47.0	329.8	3.94	13	51	177
C	5	24.9	275.0	4.98	5	14	223
D	4	31.4	290.0	3.99	6	24	212

(b) Group mean directions

	N	I (°)	D (°)	R	k	α_{95}	Palaeomagnetic pole			
							°N	°E	dp	dm
Ketilidian metavolcanics (i)	12	81.6	276.3	11.53	24	9.1	58.8	278.5	17.1	17.7
(ii)	8	69.2	306.1	7.80	35	9.5	60.4	229.8	13.8	16.2
Amphibolite dykes	4	41.6	292.0	3.76	13	27.0	31.4	214.4	20.1	32.9

N = number of samples and sites.
I = inclination of the remanence vector.
D = declination of the remanence vector.
R = magnitude of the resultant vector.
k (precision) = estimate of Fisher's κ parameter $[k = (N - I)/(N - R)]$.
α_{95} = semi-angle of the cone of 95% confidence about the mean direction.
dp and dm = semi-axes of the oval of confidence about the pole at the 95%-probability level in the colatitude direction and perpendicular to it, respectively.

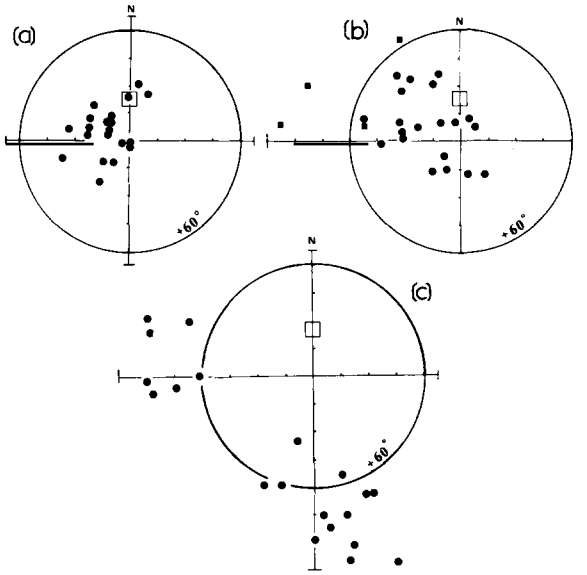


Fig. 4. Site mean directions of magnetisation: (a) of Ketilidian metavolcanics before a.f. cleaning; (b) of metavolcanics and amphibolite dykes after a.f. cleaning; and (c) after a.f. cleaning and dip correction. Polar stereographic projections; open square is the direction of the present mean geomagnetic field in this region and closed squares are amphibolite dyke directions.

tively large. Fisher's (1953) precision parameter $k = 21$ for the group mean direction from the twenty metavolcanic sites prior to dip correction but is only 4 after this correction is applied (Fig. 4). This difference is significant at the 95%-confidence level (McElhinny, 1964; Cox, 1969) and is convincing evidence that the magnetic directions isolated from these rocks postdate the second stage of regional folding. Prior to dip correction the group mean direction from the southern limb of the fold is slightly different from that derived from the western limb (Fig. 4), although this difference is not in a direction such that it could be reduced by postulating magnetisation during the course of this folding. We are unable to test whether the magnetisation predates the final stage of folding which involved folding of the cleavage imposed during the second stage.

3.1. Carriers of magnetisation

Demagnetisation characteristics of IRM's can be compared with TRM's and ARM's to examine the domain state of the magnetic mineral content (Lowrie and Fuller, 1971). In this study we find that IRM's are always less stable than TRM's and ARM's in the amphibolites (Fig.) suggesting that if magnetite is the remanence carrier then it is predominantly single domain. In the metavolcanics IRM's have comparable stabilities to TRM's (Fig. 3) suggesting that multidomain components may be present here. Johnson et al. (1974) demonstrate that the ARM stability of fine-particle magnetite (< 0.2 μm) decreases while that of coarse-particle magnetite increases with increasing field strength. For this collection ARM's are nearly constant or exhibit slight decrease in stability with increasing field strength (Figs. 2 and 3) suggesting that intermediate size ranges predominate.

This is confirmed by petrographic observation although it is not always possible to confirm that magnetite is the principle remanence carrier; pyrrhotite occurs in the Ketilidian metavolcanics and may be an important remanence carrier in these rocks. Specifically, we find magnetite exhibiting no deuteric alteration as an opaque phase in the amphibolites although at site *A* it survives as relict blebs only with primary grains pseudomorphed by hematite and ferri-rutile; hematite occurs as discrete grains (site *C*) or associated with opaque or silicate phases (sites *B–D*). The only visible difference between stable sites *A*, *C* and *D* and poorly stable site *B* is a development of maghemite around grain margins in the latter case. In the metavolcanics both the sulphides pyrrhotite and pyrite predominate usually as disseminated grains (<5 μm to limits of resolution) but occasionally as large grains, and the presence of appreciable pyrrhotite may explain the poorly stable nature of the remanence in many samples. Ilmenite and magnetite are rare as separate phases always partially or completely replaced by reddish and unresolvable alteration rims; these may represent low-grade alteration products of the primary opaque phases.

Typical thermomagnetic curves are illustrated in Fig. 5. Most samples of the metavolcanics exhibit a marked Curie point at 300–350°C, which is interpreted as due to pyrrhotite (Nagata, 1961). This mineral may retain weak ferrimagnetic properties to 550°C (Bhimasanhanam, 1964) and the residual saturation moments above this transition may be due to either magnetite or pyrrhotite. The pyrrhotite undergoes oxidation during the heating in air and a large in-

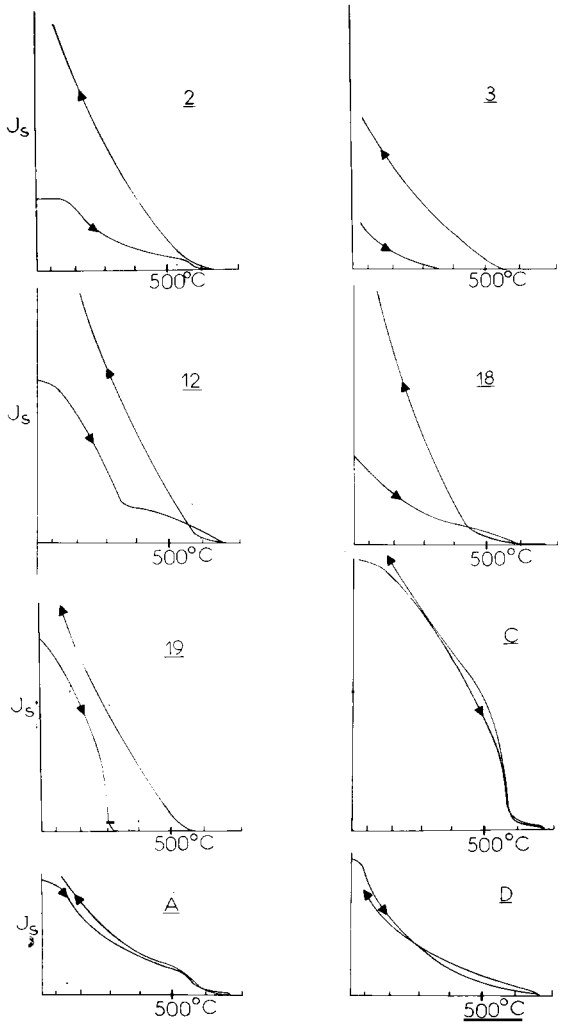


Fig. 5. Typical thermomagnetic characteristics (saturation magnetisation vs. temperature) of Ketilidian metavolcanics and amphibolite dykes.

crease in saturation remanence takes place during cooling below the Curie point of magnetite (Fig. 5). The amphibolite dykes exhibit a Curie point near that of pure magnetite at 570°C with a small residual moment which does not disappear until about 670°C, and is clearly due to hematite which has a saturation magnetisation about an order lower than magnetite (Nagata, 1961).

3.2. Pole positions

The fold test demonstrates that acquisition of magnetisations by Ketilidian metavolcanics is post-folding, and we believe that it is to be interpreted in the context of regional cooling since the directions are both widely removed from the present and from Phanerozoic field directions for North America (see McElhinny, 1973). Furthermore, magnetisation processes are enhanced with time (Néel, 1949) and when slow cooling of the crust follows regional metamorphism, magnetisations will be acquired at much lower temperatures than trapping-temperatures determined in the laboratory (Ueno et al., 1975; Morgan, 1976). Regional K–Ar studies are appropriate to dating such magnetisations because closure of common minerals to radiogenic argon, and acquisition of magnetisation take place at comparable temperatures. Thus rocks will be magnetised at temperatures as low as 200–300°C if such temperatures are maintained for periods of the order of millions of years, while minerals commonly used for K–Ar studies become closed systems to radiogenic argon at temperatures ranging from about 150 to 450°C (e.g., O'Nions et al., 1969). In the Arsůk–Ivigtut region of the Ketilidian mobile belt temperatures achieved during protracted greenschist-amphibolite facies metamorphism were thus sufficient to obliterate magnetisations held prior to folding.

The oldest pole position from this study is undoubtedly that derived from the amphibolite dykes (Fig. 6a) because intensity of regional metamorphism increases from NW–SE (Berthelesen 1960, 1962), or from the region of the thermally metamorphosed but undeformed dykes towards the Kelitidian supracrustals of Arsůk Island (Fig. 1). The duration of metamorphism was more protracted in the southwest (Bondesen and Henriksen, 1965) where major deformation took place, and we can only infer the relative ages of the poles from the Ketilidian metavolcanics from the local cooling history. It is considered that the magnetisation of the sites from the southeastern part of Arsůk Island (*13–20*) was acquired later than the sites in the west because not only are they lower in the supracrustal succession, but they occur where the regional metamorphism was most protracted (Bondesen and Henriksen, 1965; Muller, 1974).

The magnetisation of all these rocks must postdate the climax of regional metamorphism (1830 m.y.), and

TABLE III

ca. 1850–1600 m.y. palaeomagnetic pole positions before and after rotation relative to North America

Rock unit	Age (m.y.)	Palaeomagnetic pole		Pole after rotation *1		Reference
		°N	°E	°N	°E	
(i) Greenland:						
G1 Dolerite dykes, Itivdleq	1830 ± 20; 1700 ± 40	18	287	16	271	Beckmann and Mitchell (1976)
G2 Amphibolite dykes, Arsûk region	1830–1780	31	214	37	199	this paper
G3 Ketilidian metavolcanics (i)	1830–1600	59	279	58	271	this paper
G4 Ketilidian metavolcanics (ii)	1830–1600	60	230	64	221	this paper
G5 Sagdlerssuaq metamorphic rocks	1620 ± 50	57	274	57	266	Beckmann and Mitchell (1976)
G6 Itivdleq dykes and gneisses	1790–1650	–	–	13	269	Morgan (1976)
G7 Dolerite dykes between Kangâmiut and Itivdleq	–	17.1	273.8	21.5	250.9	Fahrig and Bridgwater (1976)
G8 Amphibolite boudins to the north of the Nagssugtoqidian–Archaean boundary	–	48.8	238.2	57.2	219.7	Fahrig and Bridgwater (1976)
G9 Godthåb gneisses	–	20	274	25	250	Fahrig and Bridgwater (1976)
(ii) NW Scotland:						
S1 Scourie dykes	ca. 1800 (?)	37.3	274.8	38	236	Beckmann (1976a)
(iii) Africa *2:						
A1 Mashonaland dolerites, Rhodesia	1850 ± 20 *3; 1910 ± 280 *3	7	340	−18	198	8/151
A2 Dolerite dykes, NW Sahara	<1920	35	30	29	225	8/151
A3 Waterberg Sandstones, South Africa	<1950, >1750	67	44	56	198	9/148
		41	33	35	223	9/149
		3	333	12	180	9/150
		8	10	−4	224	9/151
		36	48	39	238	9/152
A4 Van Dyk Mine, dolerite dyke, South Africa	1650	12.5	14	2	225	8/156
A5 Palabora Complex, South Africa (ii)	ca. 1900	2	9	−10	227	Briden (1976)
(iv) North America:						
N1 Dubawnt Group	1835 ± 26 *3	7	277	–	–	Park et al. (1973)
N2 Marathon dykes	1810	29	213	–	–	8/176
N3 Spanish River alkalic rocks	1790 ± 100 *3	37	264	–	–	Robertson and Watkinson (1974)
N4 Churchill Province metamorphic rocks	1622	12	259	–	–	Park (1973)
N5 Sparrow dykes	ca. 1700	12	291	–	–	McGlynn et al. (1974)
N6 Et-Then group	1845–1250	−1	312	–	–	Irving et al. (1972)
N7 Nonacho sediments	1700–1400	13	274	–	–	McGlynn et al. (1974)
N8 Western Channel Diabase	1785–1325	9	245	–	–	Irving et al. (1972)
N9 Cameron Bay Group Porphyry	1770 ± 30 *3	1	256	–	–	Irving et al. (1972)
N10 Snow Lake–Flin Flon rocks	1700–1600	24	265	–	–	Park (1975)
N11 Snow Lake–Flin Flon rocks	ca. 1800	−20	314	–	–	Park (1975)
N12 Martin Formation	1970–1650	−9	288	–	–	Evans and Bingham (1973)
N13 Stark Formation	1800–1650	–	–	–	–	Bingham and Evans (1975)

Reference numbers are listings in the *Geophysical Journal of the Royal Astronomical Society.*

*1 Using North Atlantic reconstruction (Bullard et al., 1965).

*2 Using Proterozoic supercontinent reconstruction (Piper, 1975).

*3 Rb–Sr isochrons, other ages are inferred directly or indirectly from K–Ar or other determinations.

probably also postdates post-tectonic igneous episodes (ca. 1780 m.y.) (Bridgwater et al., 1973; van Breemen et al., 1974). For reasons explained earlier, a younger age notably for the poles from the Ketilidian metavolcanics, is probable because mineral systems within the Ketilidian belt did not finally close to rubidium and argon until 1700–1600 m.y. The protracted high temperatures within the belt would have lowered the blocking temperature of the magnetic grains to a temperature comparable to that at which they became closed to these elements (Irving et al., 1974; Ueno et al., 1975) and we accordingly assign the poles to this interval.

The full potential of palaeomagnetic results from slowly-cooled metamorphic rocks has been realised by the work of Morgan (1976) who finds that virtual geomagnetic poles are distributed along a.p.w. trends both when they come from different structural (cooling) levels and when the samples possess differing coercive-force spectra. Morgan states the view that progressive demagnetisation will subtract components of higher coercive force acquired at successively higher temperatures from individual specimens; hence if apparent polar movements were taking place during regional cooling, the direction of such movements can be resolved by demagnetising at progressively higher temperatures or alternating fields. Morgan has used this principle to infer the direction of polar movement at ca. 1700 m.y. (Fig. 6a). In this collection we find that only the amphibolite-dyke samples are sufficiently stable over a wide range of demagnetising fields for them to be used to infer directions of apparent polar movement. These most stable samples do indeed show a slight but systematic movement in palaeomagnetic directions with progressive a.f. cleaning which takes place along great-circle lines close to, but significantly removed from the lines connecting the palaeomagnetic directions with the present earth's field direction (Fig. 2). We infer therefore, that this demagnetisation (except for the initial stages in some samples) is not subtracting a partial viscous remanent magnetisation acquired in the present earth's field, but rather a thermal remanent magnetisation acquired during long regional cooling. This would imply that the contemporary apparent polar movement was from the central Pacific region towards North America (Fig. 6a), but because of the uncertainties of the argument and small sample this can be regarded as a tentative conclusion only.

The results from the southern margin of the Archaean craton in west Greenland supplement two palaeomagnetic poles of Beckmann and Mitchell (1976) dated by K–Ar isochrons 1830, 1750 and 1620 m.y., the results of Morgan (1976) dated by K–Ar whole-rock ages between 1790 and 1650 m.y. (Table III), and three poles from the same region of Fahrig and Bridgwater (1976) from the northern margin of this same craton at the Nagssugtoqidian structural boundary near Holsteinsborg. Beckmann and Mitchell's result from the Sagdlerssuaq metamorphic rocks dated 1620 ± 50 m.y. by K–Ar mineral isochron agrees closely with the older pole from the Ketilidian metavolcanics (Table III) suggesting that the Ketilidian poles are indeed as young as 1700–1600 m.y.

3.3. 1850–1600 m.y. apparent polar movements

These palaeomagnetic poles from Greenland are plotted in Fig. 6 relative to North American poles (Table II) after closing the Davis Strait according to the 500-fathom continental reconstruction of Bullard et al. (1965) which involves a clockwise rotation of 18° about a Euler pole at 70.5°N, 94.4°W. The close agreement of Greenland pole *1* (K–Ar mineral isochrons 1830 and 1700 m.y.) with the pole from the Dubawnt Group (Rb–Sr isochron 1830 ± 26 m.y.) is noteworthy and would seem to confirm this as a palaeofield direction. Beckmann (1976b) favours a magnetisation during the late Laxfordian episode (1800 m.y.) for the pole from the Scourie dykes of NW Scotland, which also agrees with a pole from tonalites of the South Harris igneous complex metamorphosed in the late Laxfordian episode (Dearnley, 1963): these poles may merit comparison with poles from amphibolite dykes of this study (Fig. 6b).

The Itivdleq results described by Morgan (1976) and related to the regional cooling dated 1790–1650 m.y. by K–Ar whole-rock ages, are consistent with a number of results from the Superior Province (Table III) with assigned ages between 1790 and 1622 m.y. in addition to a pole from the Martin Formation of Saskatchewan. The Itivdleq results come from the same locality as Beckmann and Mitchell's pole with an age between 1830 and 1700 m.y. Other poles including that from the Scourie dykes (*S1*) plot further to the northwest and leave an uncertainty in the interpretation of the a.p.w. path: either the pole executed a

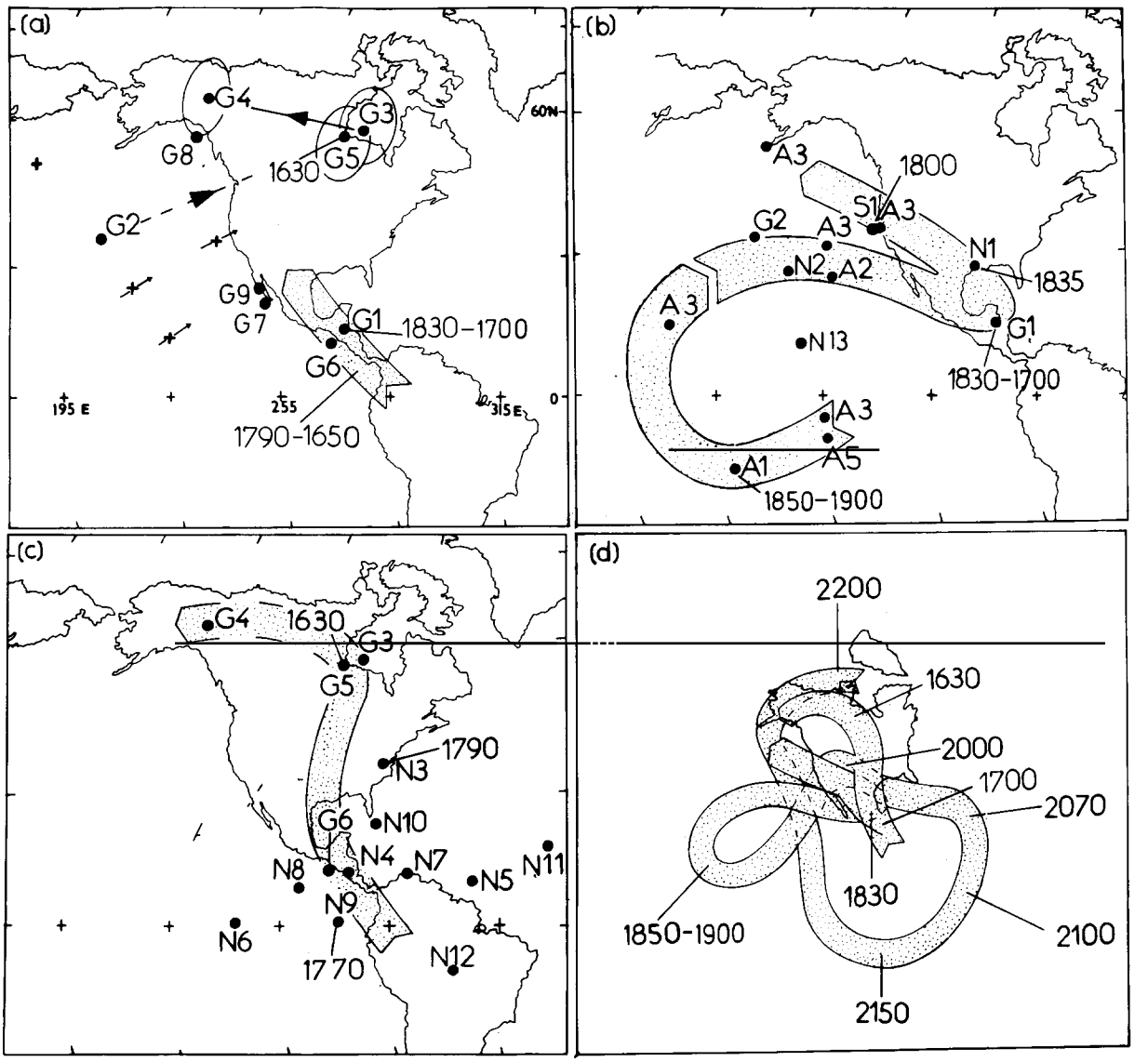


Fig. 6a. Palaeomagnetic results from the Ketilidian and Nagssugtoqidian mobile belts of Greenland and relating to the interval ca. 1850–1600 m.y. after closure of the Davis Strait. 95%-confidence ovals are given where appropriate, together with directions of apparent polar movement discussed in this paper. The a.p.w. swathe through the result *G6* is inferred by Morgan (1976) from demagnetisation behaviour of slowly cooled rocks of the Nagssugtoqidian belt, and the arrows through site virtual geomagnetic poles from amphibolite dykes (crosses) are directions of apparent polar movement suggested by demagnetisation trends of the present study.

b. Palaeomagnetic poles listed in Table III and relating approximately to the interval 1880–1800 m.y; ages considered to be of highest reliability are also indicated. A tentative polar wander path linking these poles is also indicated.

c. Palaeomagnetic poles listed in Table III and relating approximately to the interval 1800–1630 m.y. The polar wander path in this diagram connects ca. 1800–1700 m.y. results mostly from the Superior Province via the a.p.w. path recognised by Morgan (1976) (and applying to the interval 1790–1650 m.y.) to results from Greenland considered to be younger in age.

Figs. (a)–(c) use Bartholomew's equal area projection.

d. Apparent polar wander path from ca. 2200–1630 m.y. using data discussed in this paper and by McGlynn et al. (1975) and Piper (1976b).

rapid loop between 1830 m.y. and <1790 m.y. as suggested by some other poles ca. 1800 m.y. in age (Fig. 6b), Table III), or the 1830 m.y. and <1790 m.y. poles represent the same part of the a.p.w. curve and the apparent age differences are no more than a reflection of the errors associated with the assigned ages.

Also plotted in Fig. 6b are ca. 1850–1650 m.y. results from Africa using the continental reconstruction of the Proterozoic supercontinent incorporating Africa and North America which is now established by data covering the 2160–1950 m.y. (Piper, 1976b) and Upper Proterozoic times (Piper, 1975) from the Slave, Superior and Bear Tooth cratons of North America and the West Africa, Kasai, Tanganyika and Rhodesia–Kaapvaal cratons of Africa. These data provide the only current evidence for a polar loop between 1950 and 1835 m.y. but do not otherwise permit any refinement of the a.p.w. path.

It is difficult to construct a single unambiguous a.p.w. curve to embrace all of this information, but the a.p.w. curve of Fig. 6 incorporates the data from North America, Greenland and Scotland using the age sequences of successive poles discussed above together with the a.p.w. loop from Africa originally defined by Jones and McElhinny (1967) and subsequently suggested by recent North American data (pole *N13*, Fig. 6b) (Bingham and Evans, 1975); there are two large loops in the curve between 1950 and 1630 m.y. with the possibility of a third at ca. 1800 m.y. The path is more complicated than a.p.w. paths proposed hitherto from North American data alone (Spall, 1971; Irving and Park, 1972; McGlynn et al., 1975) both because the earlier curves have embraced isolated results for which the age sequence was not clear, and because some of the new results from Greenland have as yet no age equivalents from North America. The a.p.w. curve may ultimately prove to be even more complicated than shown here for two reasons. Firstly, we are recognising large polar movements within the time intervals equivalent to the age errors assigned to many of the poles, and secondly, a few results for which the assigned ages are apparently of good quality do not lie close to any contemporary poles. We do however, tentatively conclude that these discrepancies are likely to result from complexities of apparent polar movements rather than tectonic causes because of the widespread agreement of pre-1950 m.y. poles between the component cratons of North America and Africa (McGlyn et al., 1975; Piper, 1976b). It will require more detailed studies through sedimentary sequences and across metamorphic facies to resolve these a.p.w. movements in detail; the results of Morgan (1976) and this study give some insight into the potential of metamorphic rocks for solving this problem.

When the apparent movement of the pole from 2200 to 1630 m.y. (it is still poorly defined prior to 2200 m.y.) is viewed as a whole (Fig. 6d), it is clear from the curve as it is at present defined that the pole returned to similar positions at several times during this interval; other workers (e.g., Irving and Lapointe, 1975; Piper, 1976b) have noted this tendancy for the pole to return to similar positions during Proterozoic times. As more data have been forthcoming it has become clear that the polar motion is most accurately described in terms of closed loops of short duration (compare Fig. 6d) with Irving and Park (1972, fig. 2)). This observation must cast doubt on the real significance of "hairpins" or sharp changes in direction of polar movement, although it appears to remain true that periods or rapid polar movement in Proterozoic time coincide with periods of widespread tectonic activity: polar movements during the tectonically active period 2200–1700 m.y. (e.g., Sutton 1971) were very much greater than those recognised during the relatively well-studied and tectonically quiescent interval 1600–1200 m.y. [see Irving and Lapointe (1975) for summary]. Polar movements were again large after 1200 m.y. (Irving et al., 1974; Piper, 1975) during the duration of the Grenville, Sveconorwegian and related mobile episodes.

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